

Acute Effects of Transcranial Photobiomodulation On Glymphatic CSF Dynamics in Two Patients with Amnestic Mild Cognitive Impairment or Early Alzheimer Disease

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INTRODUCTION

The glymphatic system has received growing attention as a potential therapeutic target because impaired brain waste clearance is associated with neurodegenerative disorders and may represent a modifiable disease pathway [1,2]. This is particularly relevant to amnestic mild cognitive impairment and Alzheimer disease, in which amyloid- β and tau accumulation may precede overt clinical decline [3]. However, few noninvasive interventions have been shown to modulate glymphatic-relevant CSF dynamics in humans. Transcranial photobiomodulation is clinically attractive because it is noninvasive, generally well tolerated, and capable of influencing cerebral blood flow and metabolism—physiological processes linked to CSF movement [4,5]. Photobiomodulation has recently been shown to modulate CSF dynamics in healthy young adults [6], but its acute effects in cognitively impaired patients remain unknown. To our knowledge, this is the first patient-focused fMRI assessment of acute fourth-ventricular CSF responses to continuous-wave tPBM or sham in amnestic mild cognitive impairment or early Alzheimer disease.

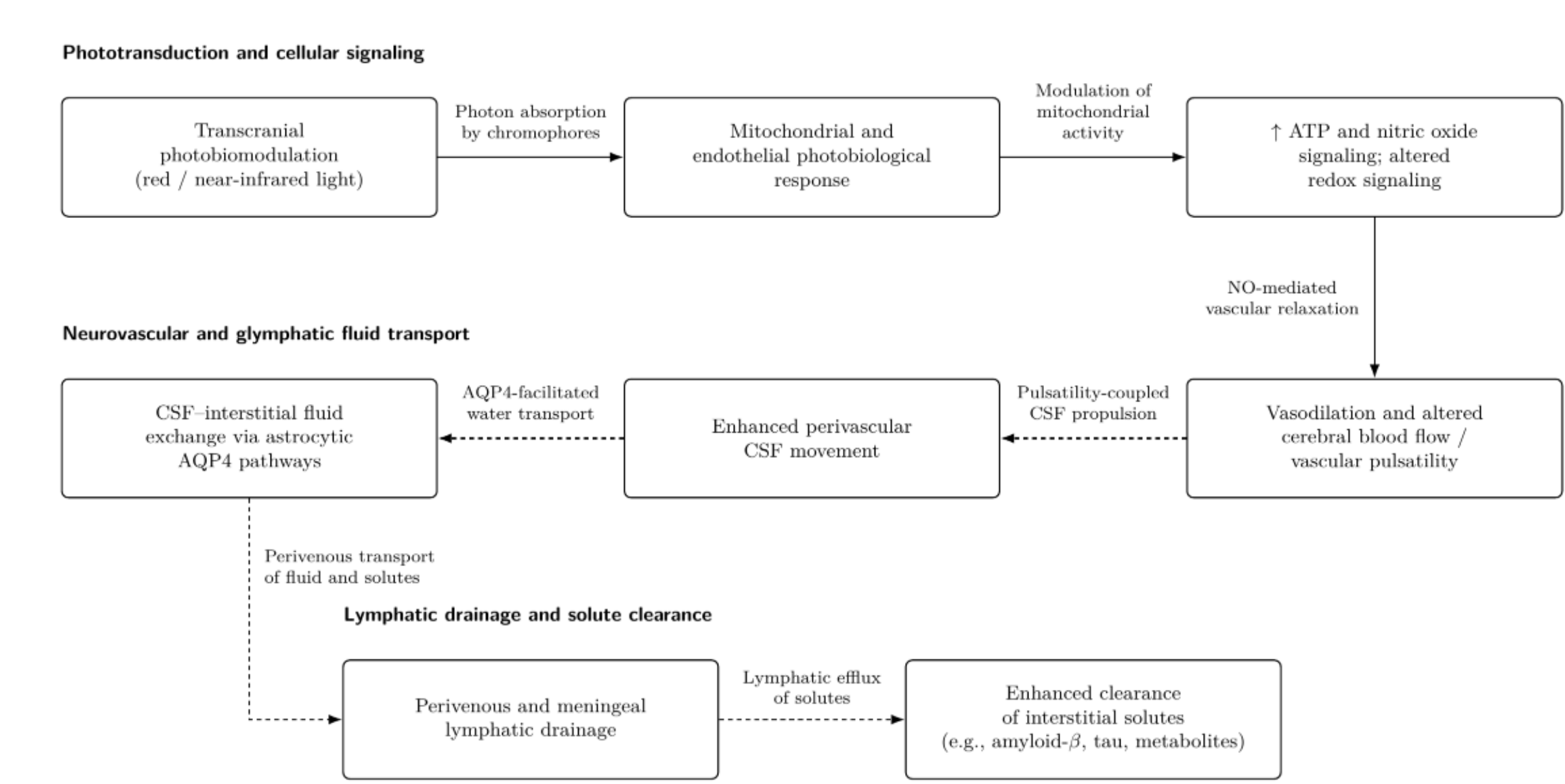


Fig. 1 Proposed Mechanism of Action

METHODS

Two participants enrolled in the TRAP-AD protocol underwent an acute fMRI session comprising approximately 10-minute acquisitions before, during, and after bilateral prefrontal 808-nm transcranial photobiomodulation. Subject 1 received continuous-wave stimulation and Subject 2 received sham. The present analysis compared the same-day pre- and post-stimulation acquisitions to evaluate fourth-ventricular CSF dynamics as an inflow-sensitive marker relevant to glymphatic function [5,6]. Multi-echo echo-planar images were acquired with a repetition time of 2.522 s; analyses used 240-volume motion-corrected second-echo data. Four-slice fourth-ventricle masks were manually defined on the native single-band reference images. Mean CSF signals were linearly detrended and expressed as percentage signal change. Time-domain fluctuation amplitude was quantified as temporal standard deviation. Power spectral density was estimated using Welch's method [8] with a 128-point Hann window and 50% overlap. Power was integrated from 0.03–0.07 Hz to quantify ultra-slow CSF oscillations centered near 0.05 Hz. The 0.198-Hz Nyquist frequency precluded assessment of cardiac-frequency pulsatility. Framewise displacement greater than 0.5 mm identified high-motion volumes [9].

REFS

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RESULTS

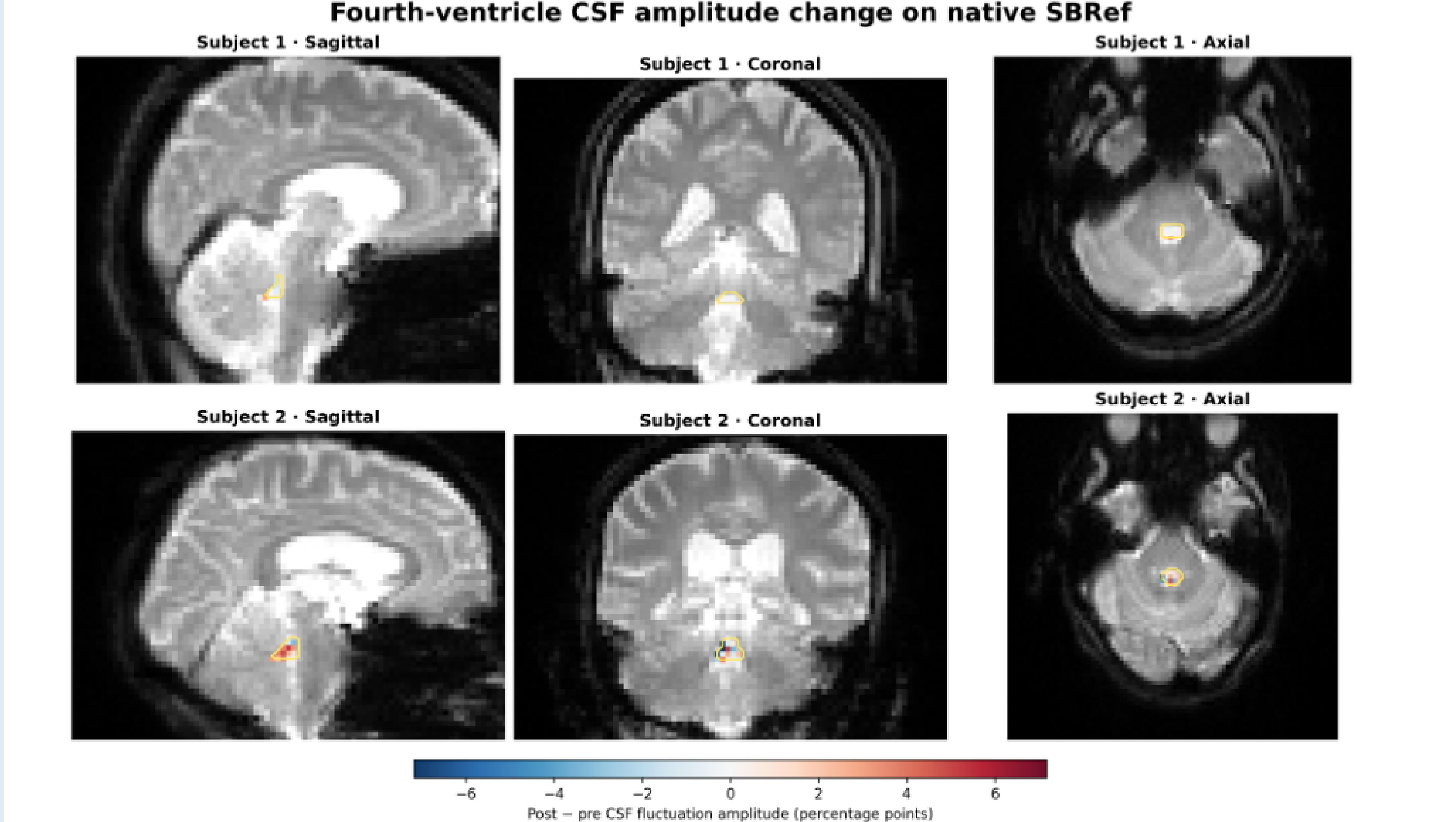


Fig. 2: Single-band Reference Acute Fourth-Ventricular CSF Changes Post-to-pre maps demonstrate relatively modest CSF amplitude changes in Subject 1 and greater regional variability in Subject 2. Warm colors indicate increased fluctuations, while cool colors indicate decreases.

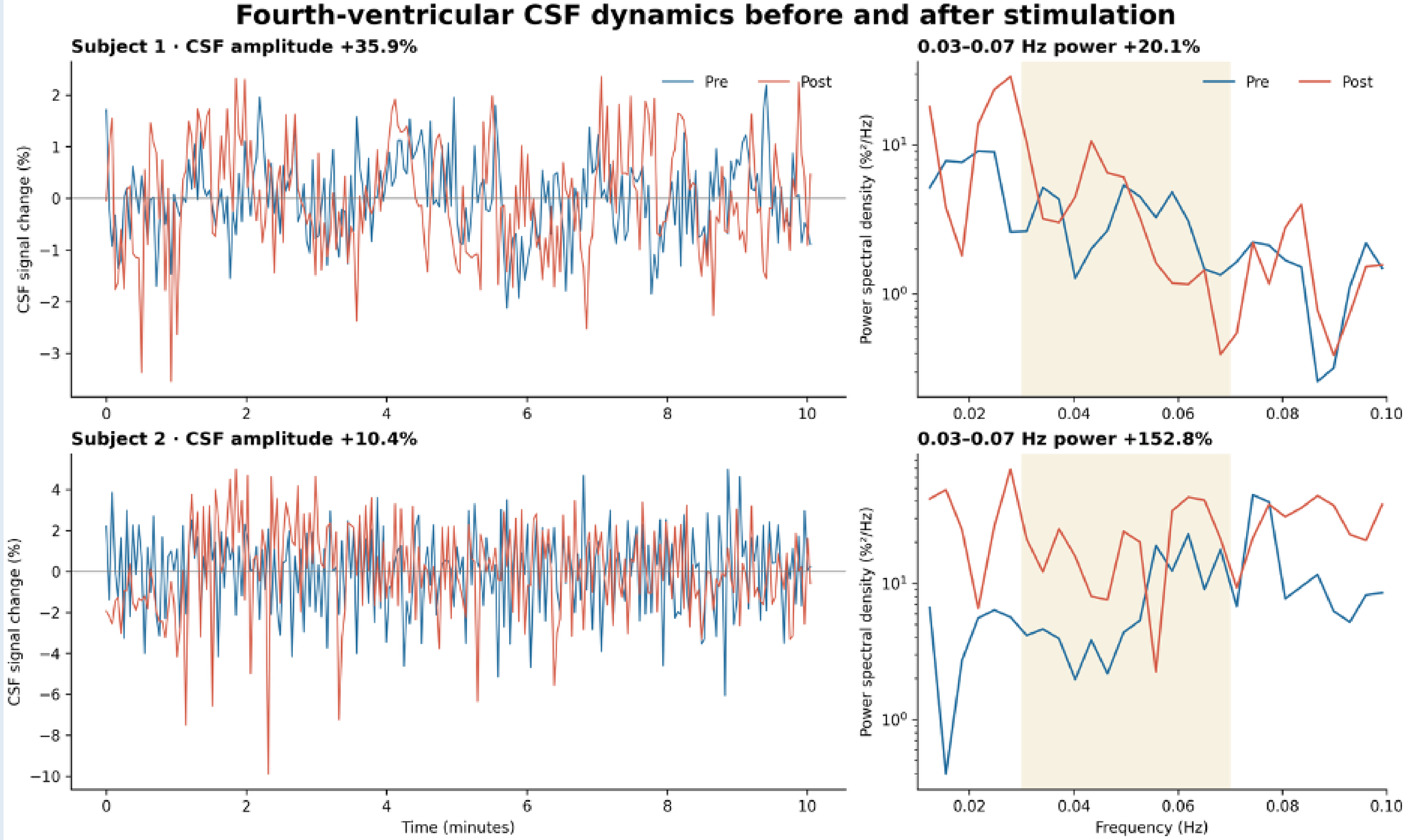


Fig. 3: Temporal and Ultra-Slow Spectral Changes in Fourth-Ventricular CSF Dynamics Following continuous-wave tPBM, Subject 1 demonstrated a 35.9% increase in CSF fluctuation amplitude and a 20.1% increase in 0.03–0.07 Hz power. Following sham, Subject 2 demonstrated increases of 10.4% and 152.8%, respectively.

In Subject 1, CSF fluctuation amplitude increased from 0.792% before continuous-wave stimulation to 1.077% afterward, a 35.9% increase. Ultra-slow 0.03–0.07 Hz power increased from 0.124 to 0.148 %², representing a 20.1% increase. However, this band's proportion of total 0.01–0.10 Hz power decreased from 42.3% to 32.2%, suggesting that the post-stimulation increase was not confined to the approximately 0.05-Hz range. In Subject 2, CSF amplitude increased from 2.089% before sham to 2.307% afterward, a 10.4% increase. Ultra-slow power increased from 0.310 to 0.785 %², a 152.8% increase, while its proportion of total slow-frequency power decreased from 36.0% to 33.8%. Subject 2 had 8 of 240 high-motion volumes before sham and 76 of 240 afterward. Consequently, the large post-sham spectral increase likely contains substantial motion-related broadband power.

DISCUSSION AND CONCLUSION

Patient 1, who received continuous-wave tPBM, demonstrated concordant increases in fourth-ventricular CSF fluctuation amplitude and ultra-slow oscillatory power, consistent with acute modulation of glymphatic-relevant CSF dynamics. Patient 2, who received sham, showed a smaller increase in CSF amplitude but a larger spectral-power increase; however, substantial post-sham motion limits interpretation of this finding [9]. Although inflow-sensitive fMRI does not directly quantify glymphatic drainage or waste clearance [5,6], the Patient 1 response provides initial clinical evidence that tPBM may engage a CSF component of glymphatic physiology.

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